

REPAIRED-GAUGE CONSTRUCTION FOR SINGLE-CORE TYPE II NAVIER-STOKES CONCENTRATION

ABSTRACT. This paper gives the local representation theorem for the retained nondegenerate single-core Type II stratum. From explicit physical-core assumptions and the nondegenerate repaired-gauge selection defining this stratum on a compact time window we obtain:

- (1) a preliminary scale-center path and its renormalized map,
- (2) exact transformed equations with fixed sign conventions,
- (3) an explicit finite-dimensional gauge map,
- (4) a full Jacobian computation and Schur-complement invertibility criterion,
- (5) propagation of the repaired-gauge constraints along the absolutely continuous modulation path,
- (6) exact local pressure transport and a local pressure decomposition,
- (7) explicit modulation coefficients from differentiated constraints, and
- (8) preservation of local suitable structure in renormalized variables.

Failure of this local repaired-gauge selection is not treated as a defect of the single-core representation theorem. It is the gauge-degenerate/multibubble or cascade alternative in the state-space stratification. On the retained nondegenerate stratum, pressure transport, modulation formulas, suitable structure, and compact-window representation follow by the theorems below.

1. ASSUMPTIONS AND NOTATION

Let $u : \mathbb{R}^3 \times (T - \delta, T) \rightarrow \mathbb{R}^3$, $p : \mathbb{R}^3 \times (T - \delta, T) \rightarrow \mathbb{R}$ be a suitable weak Navier–Stokes solution with viscosity $\nu > 0$. For $z = (x_0, t_0)$ and $R > 0$,

$$Q_R(z) := B_R(x_0) \times (t_0 - R^2, t_0), \quad Q_R^* := B_R \times I_*.$$

Fix a reference physical center-time $(x_\bullet, t_\star)$ around which $B_{R_\star}(x_\bullet)$ is taken. For a fixed spatial center x_0 we also use

$$Q_R(t) := Q_R(x_0, t).$$

Fix a compact window $I_* := [\tau_0 - \ell, \tau_0 + \ell]$, $\ell > 0$, and write $I_{\text{ph}} := [t_\star - \Lambda_{\max}^2 \ell, t_\star + \Lambda_{\max}^2 \ell]$.

For a cylinder $Q_R(x_0, t_0) = B_R(x_0) \times (t_0 - R^2, t_0)$, set

$$\begin{aligned} A(u; Q_R(x_0, t_0)) &:= R^{-1} \sup_{s \in (t_0 - R^2, t_0)} \int_{B_R(x_0)} |u(x, s)|^2 dx, \\ E(u; Q_R(x_0, t_0)) &:= R^{-1} \int_{Q_R(x_0, t_0)} |\nabla u|^2 dx ds, \\ C(u; Q_R(x_0, t_0)) &:= R^{-2} \int_{Q_R(x_0, t_0)} |u|^3 dx ds, \\ D(p; Q_R(x_0, t_0)) &:= R^{-2} \int_{t_0 - R^2}^{t_0} \int_{B_R(x_0)} |p(x, s) - \langle p(s) \rangle_{B_R(x_0)}|^{3/2} dx ds. \end{aligned}$$

Here $\langle p(s) \rangle_{B_R(x_0)} := |B_R|^{-1} \int_{B_R(x_0)} p(x, s) dx$; pressure is always understood modulo time-dependent constants.

Definition 1.1 (Single-core assumption package). Let $(R_\star, r_\star, M, \eta, \kappa, \Lambda_{\min}, \Lambda_{\max})$ satisfy $\Lambda_{\min} \leq \Lambda_{\max} < \infty$. We say (u, p) satisfies SC($R_\star, r_\star, M, \eta, \kappa$) on $Q_{R_\star}(x_\bullet, t_\star)$ if the following hold:

(SC1) $A(u; Q_{R_*}(x_\bullet, t)) + E(u; Q_{R_*}(x_\bullet, t)) + C(u; Q_{R_*}(x_\bullet, t)) + D(p; Q_{R_*}(x_\bullet, t)) \leq M$ for every $t \in [t_* - R_*^2, t_*]$.

(SC2) There exists a physically selected core map $(x_c(t), \rho_c(t))$ on I_{ph} such that $\rho_c(t) \in (0, R_*]$, $x_c(t) \in B_{R_*/2}(x_\bullet)$, and for some $\Theta_0 > 0$,

$$\int_{B_{r_*}(0)} |\rho_c(t)^{-1} u(x_c(t) + \rho_c(t)y, t)|^3 dy \geq \eta \quad \text{for a.e. } t \in I_{\text{ph}}.$$

(SC3) There exists $0 < r_{\text{sep}} \leq r_*/2$ such that for every $0 < r \leq r_{\text{sep}}$, every competitor pair

$$\mathcal{C}_{\text{core}}(t) := \left\{ (\bar{x}, \bar{\rho}) : |\bar{x} - x_\bullet| \leq \frac{R_*}{2}, \ 0 < \bar{\rho} \leq R_* \right\},$$

$(\bar{x}, \bar{\rho}) \in \mathcal{C}_{\text{core}}(t) \setminus \{(x_c(t), \rho_c(t))\}$ with

$$\rho_c(t)^{-1} |\bar{x} - x_c(t)| + \left| \frac{\bar{\rho}}{\rho_c(t)} - 1 \right| \geq r_{\text{sep}}$$

has separated local weighted mass in the same core-scale variables:

$$\left| \int_{B_r} |\rho_c(t)^{-1} u(x_c(t) + \rho_c(t)y, t)|^3 dy - \int_{B_r} |\bar{\rho}^{-1} u(\bar{x} + \bar{\rho}y, t)|^3 dy \right| \geq \eta.$$

This is the quantitative non-competing core condition, and is imposed on the selected physical core directly; it is not a consequence of the repaired-gauge construction.

(SC4) For the selected core,

$$\sup_{t \in I_{\text{ph}}} \int_{B_{2R_*}} |y|^{-p} |\rho_c(t)^{-1} u(x_c(t) + \rho_c(t)y, t)|^3 dy \leq M,$$

for some $0 < p < 3$.

(SC5) The preliminary frame associated with the selected core is the absolutely continuous representative

$$0 < \Lambda_{\min} \leq \Lambda_{\text{pre}}(\tau) \leq \Lambda_{\max}, \quad \Lambda_{\text{pre}} \in AC(I_*), \quad X_{\text{pre}} \in AC(I_*; \mathbb{R}^3).$$

It is tied to the selected physical core by the time change $dt_c/d\tau = \Lambda_{\text{pre}}(\tau)^2$, $t_c(\tau_0) = t_*$: one has

$$X_{\text{pre}}(\tau) = x_c(t_c(\tau)), \quad \Lambda_{\text{pre}}(\tau) = \rho_c(t_c(\tau))$$

for a.e. $\tau \in I_*$.

The parameter κ records the quantitative constants obtained in the local gauge estimates and is not used in items (SC1)–(SC5).

Remark 1.2. The selected core (x_c, ρ_c) and its separation/nondegeneracy conditions are specified in physical variables; no repaired-gauge constraint is used in their definition. This avoids the circularity of defining the core via the chart to be constructed.

2. PRELIMINARY FRAME MAP FROM THE SELECTED CORE

This section records the transport map generated by the selected core; no new core extraction is proved here.

Define

$$\frac{dt_c}{d\tau} = \Lambda_{\text{pre}}(\tau)^2, \quad t_c(\tau_0) = t_*,$$

and use the compatibility in (SC5) to identify $\Lambda_{\text{pre}}(\tau) = \rho_c(t_c(\tau))$ and $X_{\text{pre}}(\tau) = x_c(t_c(\tau))$ a.e.

Proposition 2.1 (Preliminary frame map regularity). *Under $\text{SC}(R_*, r_*, M, \eta, \kappa)$, the map*

$$\Phi_{\text{pre}}(\tau, Y) := (x, t) = (X_{\text{pre}}(\tau) + \Lambda_{\text{pre}}(\tau)Y, t_c(\tau))$$

is well-defined and absolutely continuous in τ . Moreover, $1/\Lambda_{\text{pre}}, \Lambda_{\text{pre}} \in L^\infty(I_)$, $X_{\text{pre}} \in W_{\text{loc}}^{1,1}(I_*; \mathbb{R}^3)$, and for every $R \leq R_*$,*

$$\Phi_{\text{pre}} : Q_R^* \rightarrow \Phi_{\text{pre}}(Q_R^*) \subset Q_{2R_*}(x_\bullet, t_\star)$$

is bi-Lipschitz with $\det \nabla_Y \Phi_{\text{pre}} = \Lambda_{\text{pre}}^3$, hence Jacobian and inverse-Jacobian bounds depend only on $\Lambda_{\min}, \Lambda_{\max}$.

Proof. From (SC5), $\Lambda_{\text{pre}} \in AC(I_*)$, $0 < \Lambda_{\min} \leq \Lambda_{\text{pre}} \leq \Lambda_{\max} < \infty$, and $X_{\text{pre}} \in AC(I_*; \mathbb{R}^3)$. The scalar ODE

$$\frac{dt_c}{d\tau} = \Lambda_{\text{pre}}(\tau)^2, \quad t_c(\tau_0) = t_\star$$

therefore has an absolutely continuous strictly increasing solution on I_* , and

$$\frac{dt_c}{d\tau} = \Lambda_{\text{pre}}(\tau)^2 \quad \text{a.e. on } I_*.$$

The compatibility condition (SC5) identifies this absolutely continuous frame with the selected physical core a.e.; no differentiability of x_c or ρ_c beyond the frame regularity in (SC5) is used. For fixed τ , $\nabla_Y \Phi_{\text{pre}}(\tau, \cdot) = \Lambda_{\text{pre}}(\tau)I$, so

$$\det \nabla_Y \Phi_{\text{pre}}(\tau, \cdot) = \Lambda_{\text{pre}}(\tau)^3, \quad \Lambda_{\min}^3 \leq \det \nabla_Y \Phi_{\text{pre}} \leq \Lambda_{\max}^3.$$

Hence, if $Y_1, Y_2 \in B_R$,

$$\Lambda_{\min}|Y_1 - Y_2| \leq |\Phi_{\text{pre}}(\tau, Y_1) - \Phi_{\text{pre}}(\tau, Y_2)| \leq \Lambda_{\max}|Y_1 - Y_2|,$$

and therefore $\Phi_{\text{pre}}(\tau, \cdot)$ is bi-Lipschitz on B_R . Its inverse has Lipschitz constant $\Lambda_{\min}^{-1}$, so Jacobian and inverse-Jacobian bounds are controlled by $\Lambda_{\min}, \Lambda_{\max}$.

For image inclusions, if $x = X_{\text{pre}}(\tau) + \Lambda_{\text{pre}}(\tau)Y$ with $|Y| \leq R$, then

$$|x - X_{\text{pre}}(\tau)| \leq \Lambda_{\max}R, \quad X_{\text{pre}}(\tau) \in B_{R_*/2}(x_\bullet),$$

so $x \in B_{R_*/2+\Lambda_{\max}R}(x_\bullet)$. As $\tau \in I_*$, we also have $t = t_c(\tau) \in I_{\text{ph}}$. Therefore

$$\Phi_{\text{pre}}(Q_R^*) \subset B_{R_*/2+\Lambda_{\max}R}(x_\bullet) \times I_{\text{ph}}.$$

□

Definition 2.2 (Preliminary profile). For $(x, t) = \Phi_{\text{pre}}(\tau, Y)$, define

$$V_{\text{pre}}(Y, \tau) := \Lambda_{\text{pre}}(\tau) u(x, t), \quad P_{\text{pre}}(Y, \tau) := \Lambda_{\text{pre}}(\tau)^2 p(x, t).$$

Hence $u(x, t) = \Lambda_{\text{pre}}(\tau)^{-1}V_{\text{pre}}(Y, \tau)$ and $p(x, t) = \Lambda_{\text{pre}}(\tau)^{-2}P_{\text{pre}}(Y, \tau)$ under Φ_{pre} .

3. EXACT RENORMALIZED CHANGE OF VARIABLES

Define renormalized fields on Q_{2R}^* by

$$Y = \frac{x - X(\tau)}{\Lambda(\tau)}, \quad u(x, t) = \Lambda(\tau)^{-1}V(Y, \tau), \quad p(x, t) = \Lambda(\tau)^{-2}P(Y, \tau),$$

where $dt/d\tau = \Lambda(\tau)^2$, $X : I_* \rightarrow \mathbb{R}^3$, $\Lambda : I_* \rightarrow (0, \infty)$, and $\Lambda, X \in AC(I_*)$. Write

$$a(\tau) := -\Lambda'(\tau), \quad b(\tau) := -X'(\tau).$$

Lemma 3.1 (Exact differential identities). *For smooth data and fixed (x, t) , the chain rule gives*

$$\begin{aligned}\partial_t u &= \Lambda^{-3} (\partial_\tau V - (X' + \Lambda' Y) \cdot \nabla_Y V - \Lambda' V), \\ \nabla_x u &= \Lambda^{-2} \nabla_Y V, \quad \Delta_x u = \Lambda^{-3} \Delta_Y V, \quad \nabla_x p = \Lambda^{-3} \nabla_Y P.\end{aligned}$$

In particular

$$\nabla_x \cdot u = \Lambda^{-2} \nabla_Y \cdot V.$$

Proof. For fixed (x, t) , $\tau = \tau(t)$ and $Y = (x - X(\tau))/\Lambda(\tau)$. Differentiate

$$u(x, t) = \Lambda(\tau)^{-1} V(Y, \tau)$$

in t :

$$\partial_t u = -\frac{\Lambda'}{\Lambda^2} V + \frac{1}{\Lambda} \partial_\tau V \frac{d\tau}{dt} + \frac{1}{\Lambda} \nabla_Y V \cdot \frac{dY}{dt}.$$

Since $d\tau/dt = \Lambda^{-2}$ and

$$Y_\tau = -\Lambda^{-1} X' - \frac{\Lambda'}{\Lambda} Y, \quad \frac{dY}{dt} = Y_\tau \frac{d\tau}{dt},$$

we have

$$\partial_t u = \Lambda^{-3} (\partial_\tau V - (X' + \Lambda' Y) \cdot \nabla_Y V - \Lambda' V).$$

For spatial derivatives, $Y = (x - X)/\Lambda$ gives $\nabla_x Y = \Lambda^{-1} I$, hence

$$\nabla_x u = \Lambda^{-2} \nabla_Y V, \quad \Delta_x u = \Lambda^{-3} \Delta_Y V.$$

Similarly $p(x, t) = \Lambda^{-2} P(Y, \tau)$ yields

$$\nabla_x p = \Lambda^{-3} \nabla_Y P.$$

Taking divergence of $u = \Lambda^{-1} V$ gives

$$\nabla_x \cdot u = \Lambda^{-2} \nabla_Y \cdot V.$$

□

Theorem 3.2 (Renormalized Navier–Stokes equation). *Assume u, p solve*

$$\partial_t u + (u \cdot \nabla_x) u + \nabla_x p - \nu \Delta_x u = 0, \quad \nabla_x \cdot u = 0$$

in distributions. Under the transformation above, V, P satisfy, on the image renormalized cylinder, the exact equation

$$\partial_\tau V + (V \cdot \nabla) V + \nabla P - \nu \Delta V + a(\tau)(V + Y \cdot \nabla V) + b(\tau) \cdot \nabla V = 0, \quad \nabla \cdot V = 0,$$

with the sign convention fixed by $a = -\Lambda'$, $b = -X'$. Conversely, any V, P satisfying this equation pull back to a distributional solution u, p of Navier–Stokes on the physical image.

Proof. To avoid hidden regularity loss, first write the physical equation tested against $\Phi \in C_c^\infty$ in the physical cylinder and then use the smooth pull-back formula (an approximation argument gives distributions). By Lemma 3.1,

$$\partial_t u + (u \cdot \nabla_x) u + \nabla_x p - \nu \Delta_x u = 0$$

becomes

$$\Lambda^{-3} (\partial_\tau V - (X' + \Lambda' Y) \cdot \nabla V - \Lambda' V) + \Lambda^{-3} (V \cdot \nabla) V + \Lambda^{-3} \nabla P - \Lambda^{-3} \nu \Delta V = 0.$$

Multiplying by Λ^3 and using $a = -\Lambda'$, $b = -X'$ gives

$$\partial_\tau V + (V \cdot \nabla) V + \nabla P - \nu \Delta V + a(V + Y \cdot \nabla V) + b \cdot \nabla V = 0.$$

The divergence condition is

$$\nabla_x \cdot u = \Lambda^{-2} \nabla \cdot V = 0 \implies \nabla \cdot V = 0.$$

Conversely, assume (V, P) solve this equation on the renormalized cylinder. Set u, p by the same change-of-variables map. Then the preceding identities run in reverse and give

$$\partial_t u + (u \cdot \nabla_x)u + \nabla_x p - \nu \Delta_x u = 0, \quad \nabla_x \cdot u = 0$$

in $\mathcal{D}'$ on the physical image. $\square$

4. DOMAIN BOOKKEEPING AND SUPPORT

Let $R > 0$ be fixed and $0 < \ell \leq \ell_0$. Set $Q_R^* = B_R \times I_*$. With Φ as above define physical support maps

$$\mathcal{Q}_R^{\text{ph}} := \Phi(Q_R^*), \quad \mathcal{Q}_{2R}^{\text{ph}} := \Phi(Q_{2R}^*).$$

Proposition 4.1 (Compact support transport). *Assume $\Lambda_{\min} \leq \Lambda \leq \Lambda_{\max}$ on I_* . Then*

$$X(\tau) \in B_{R^*/2}(x_\bullet), \quad |x - X(\tau)| \leq \Lambda_{\max} R,$$

$$\mathcal{Q}_R^{\text{ph}} \subset B_{R^*/2 + \Lambda_{\max} R}(x_\bullet) \times I_{\text{ph}},$$

$$\mathcal{Q}_{2R}^{\text{ph}} \subset B_{R^*/2 + 2\Lambda_{\max} R}(x_\bullet) \times I_{\text{ph}},$$

for every $\tau \in I_*$ and $x = X(\tau) + \Lambda(\tau)Y$ with $Y \in B_{2R}$. In particular, if $|c(\tau)| \leq \delta$ from Proposition 7.4, then $X(\tau) \in B_{R^*/2 + \Lambda_{\max} \delta}(x_\bullet)$ on I_* . Hence, $\chi_R \in C_c^\infty(B_{2R})$ pulls back to $\chi_R^\tau := \chi_R \circ \Phi^{-1} \in C_c^\infty(\mathcal{Q}_{2R}^{\text{ph}})$ with support depending only on $R, \Lambda_{\min}, \Lambda_{\max}$.

Proof. Fix $\tau \in I_*$ and $Y \in B_R$. Since $x = X(\tau) + \Lambda(\tau)Y$,

$$|x - X(\tau)| \leq |Y| \Lambda_{\max} \leq \Lambda_{\max} R, \quad \Lambda_{\max}^{-1} |x - X(\tau)| \leq |Y|.$$

Because $X(\tau) \in B_{R^*/2}(x_\bullet)$, this gives

$$x \in B_{R^*/2 + \Lambda_{\max} R}(x_\bullet).$$

Similarly, if $Y \in B_{2R}$, then $x \in B_{R^*/2 + 2\Lambda_{\max} R}(x_\bullet)$. Together with $t = t_c(\tau) \in I_{\text{ph}}$, these give the claimed inclusions for $\mathcal{Q}_R^{\text{ph}}$ and $\mathcal{Q}_{2R}^{\text{ph}}$.

For pull-backs of cutoffs, Φ is bi-Lipschitz at every τ , so composition

$$\chi_R^\tau(x, t) := \chi_R\left(\frac{x - X(\tau)}{\Lambda(\tau)}\right)$$

is smooth on $\mathcal{Q}_{2R}^{\text{ph}}$, supported where $Y \in B_{2R}$, hence supported where $(x, t) \in \mathcal{Q}_{2R}^{\text{ph}}$. The support size depends only on $R, \Lambda_{\min}, \Lambda_{\max}$ and the fixed C_c^∞ norm of χ_R , not on the particular branch. $\square$

5. TRANSPORT OF LOCAL QUANTITIES UNDER THE CHART

Let $t = t(\tau) = t_c(\tau)$ and, for $R > 0$, define

$$J_\tau^-(R) := (t(\tau) - \Lambda_{\max}^2 R^2, t(\tau)].$$

Define renormalized local quantities on $Q_R^*(\tau)$ by

$$\begin{aligned} A_V(\tau, R) &:= \sup_{s \in (\tau - R^2, \tau)} \int_{B_R} |V(\cdot, s)|^2 dY, & E_V(\tau, R) &:= \int_{\tau - R^2}^\tau \int_{B_R} |\nabla V|^2 dY ds, \\ C_V(\tau, R) &:= \int_{\tau - R^2}^\tau \int_{B_R} |V|^3 dY ds, & D_V(\tau, R) &:= \int_{\tau - R^2}^\tau \int_{B_R} \left| P - \langle P \rangle_{B_R} \right|^{3/2} dY ds. \end{aligned}$$

For interval notation and a center X , define

$$A_u(J, R; X) := \sup_{s \in J} \int_{B_R(X)} |u(\cdot, s)|^2 dx, \quad E_u(J, R; X) := \int_J \int_{B_R(X)} |\nabla u|^2 dx ds,$$

$$C_u(J, R; X) := \int_J \int_{B_R(X)} |u|^3 dx ds, \quad D_u(J, R; X) := \int_J \int_{B_R(X)} |p - \langle p \rangle_{B_R(X)}|^{3/2} dx ds.$$

In the inequalities below we abbreviate

$$\begin{aligned} A_u(J, R) &:= A_u(J, R; X(\tau)), & E_u(J, R) &:= E_u(J, R; X(\tau)), \\ C_u(J, R) &:= C_u(J, R; X(\tau)), & D_u(J, R) &:= D_u(J, R; X(\tau)). \end{aligned}$$

Proposition 5.1 (Cylinder comparison on compact windows). *Assume $\Lambda_{\min} \leq \Lambda \leq \Lambda_{\max}$ and formulas of Lemma 3.1. Then for every $\tau \in I_*$, $R > 0$,*

$$\begin{aligned} A_V(\tau, R) &\leq \Lambda_{\min}^{-1} A_u(J_\tau^-(R), \Lambda_{\max} R), \\ E_V(\tau, R) &\leq \Lambda_{\min}^{-3} E_u(J_\tau^-(R), \Lambda_{\max} R), \\ C_V(\tau, R) &\leq \Lambda_{\min}^{-2} C_u(J_\tau^-(R), \Lambda_{\max} R), \\ D_V(\tau, R) &\leq \Lambda_{\min}^{-2} D_u(J_\tau^-(R), \Lambda_{\max} R). \end{aligned}$$

In particular, all four renormalized quantities are bounded whenever the physical quantities are bounded on the corresponding compact physical cylinders.

Proof. These are direct changes of variables under Φ_{pre} :

$$\begin{aligned} V(\cdot, \tau) &= \Lambda(\tau) u(\cdot, t(\tau)), & P(\cdot, \tau) &= \Lambda(\tau)^2 p(\cdot, t(\tau)), \\ \nabla_Y V &= \Lambda \nabla_x u, & dY d\tau &= \Lambda^{-5} dx dt, & d\tau &= \Lambda^2 dt. \end{aligned}$$

Insert these formulas into each definition and use $\Lambda_{\min} \leq \Lambda \leq \Lambda_{\max}$, plus

$$\tau - s \in (0, R^2) \implies t(\tau) - \Lambda_{\max}^2 R^2 \leq t(s) \leq t(\tau) - \Lambda_{\min}^2 R^2.$$

□

6. LOCAL REPAIRED GAUGE MAP

Fix $R > 0$, $0 < p < 3$, and $\chi_R \in C_c^\infty(B_{2R})$, $\chi_R \equiv 1$ on B_R . Let the local energy profile space be

$$\mathcal{V}_R := \left\{ V \in L^3(B_{2R}) \cap W^{1,3/2}(B_{2R}; \mathbb{R}^3) : |Y|^{-p/3} V \in L^3(B_{2R}) \right\},$$

with norm

$$\|V\|_{\mathcal{V}_R} := \|V\|_{L^3(B_{2R})} + \| |Y|^{-p/3} V \|_{L^3(B_{2R})} + \|\nabla V\|_{L^{3/2}(B_{2R})}.$$

For differentiating the repaired scale and centering gauges we use the admissible gauge class

$$\mathcal{A}_{p,R} := \left\{ V : \begin{aligned} &V \in L^3(B_{4R}) \cap W_{\text{loc}}^{1,3}(B_{4R}; \mathbb{R}^3), \\ &\int_{B_{4R}} |Y|^{-p} |V|^3 dY < \infty, \quad \int_{B_{4R}} |Y|^{-p} |V| |Y \cdot \nabla V| dY < \infty, \\ &\int_{B_{4R}} |Y|^{-p-1} |V|^3 dY < \infty \end{aligned} \right\}.$$

The larger ball is only a domain buffer: all gauge integrals are supported in B_{2R} , while (λ, c) is restricted near $(1, 0)$ so that $\lambda^{-1}(B_{2R} - c) \subset B_{4R}$.

Definition 6.1 (Gauge actions). For $(\lambda, c) \in (0, \infty) \times \mathbb{R}^3$, define

$$V_{\lambda,c}(Y) := \lambda^{-1} V(\lambda^{-1}(Y - c)).$$

For pressure profiles use

$$P_{\lambda,c}(Y) := \lambda^{-2} P(\lambda^{-1}(Y - c)).$$

We also write $V^{\lambda,c} := V_{\lambda,c}$ and $P^{\lambda,c} := P_{\lambda,c}$.

Definition 6.2 (Finite-dimensional gauge map). For $V \in \mathcal{V}_R$ and $(\lambda, c) \in (0, \infty) \times \mathbb{R}^3$, set

$$\mathcal{F}(\lambda, c; V) := (\mathcal{F}_0, \mathcal{F}_1, \mathcal{F}_2, \mathcal{F}_3),$$

with

$$\mathcal{F}_0(\lambda, c; V) := \int_{\mathbb{R}^3} \chi_R(Y) |Y|^{-p} |V_{\lambda, c}(Y)|^3 dY - \Theta_0, \quad \mathcal{F}_j(\lambda, c; V) := \int_{\mathbb{R}^3} Y_j \chi_R(Y) |V_{\lambda, c}(Y)|^2 dY$$

for $j = 1, 2, 3$. The gauge conditions are $\mathcal{F}(\lambda, c; V) = 0$. Codomain is $\mathbb{R}^4$, domain $(0, \infty) \times \mathbb{R}^3 \times \mathcal{V}_R$.

Definition 6.3 (Localized modulation matrix). For $J \subset I_*$ and a profile path $W : J \rightarrow \mathcal{V}_R$, define

$$J_W(\tau) := D_{(\lambda, c)} \mathcal{F}(1, 0; W(\tau)) \in \mathbb{R}^{4 \times 4}.$$

The rows are ordered as the scale constraint followed by the three centering constraints.

Lemma 6.4 (Weighted integrability and differentiation under integrals). Assume $V, W \in \mathcal{A}_{p, R}$. Then $G_0(V) := \int \chi_R |Y|^{-p} |V|^3$ is finite and

$$\left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \int \chi_R |Y|^{-p} |V + \epsilon W|^3 dY = \int 3\chi_R |Y|^{-p} |V| V \cdot W dY.$$

Proof. For $W \in \mathcal{A}_{p, R}$, define

$$F_\epsilon(Y) := \chi_R(Y) |Y|^{-p} |V(Y) + \epsilon W(Y)|^3.$$

For each Y , $\epsilon \mapsto F_\epsilon(Y)$ is C^1 and

$$\partial_\epsilon F_\epsilon = 3\chi_R |Y|^{-p} |V + \epsilon W| (V + \epsilon W) \cdot W.$$

For $|\epsilon| \leq 1$,

$$|F_\epsilon(Y)| + |\partial_\epsilon F_\epsilon| \leq C \chi_R |Y|^{-p} (|V|^3 + |W|^3) \in L^1(B_{2R})$$

by the defining weighted L^3 condition of $\mathcal{A}_{p, R}$. Hence dominated convergence allows differentiation under the integral, and

$$\left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \int_{B_{2R}} \chi_R |Y|^{-p} |V + \epsilon W|^3 dY = \int_{B_{2R}} 3\chi_R |Y|^{-p} |V| V \cdot W dY.$$

□

Lemma 6.5 (Gauge covariance). For $V \in \mathcal{V}_R$, $(\lambda, c) \in (0, \infty) \times \mathbb{R}^3$, and $\alpha = 0, \dots, 3$,

$$\mathcal{F}_\alpha(1, 0; V_{\lambda, c}) = \mathcal{F}_\alpha(\lambda, c; V).$$

Proof. For each component this is a direct substitution:

$$\mathcal{F}_0(1, 0; V_{\lambda, c}) = \int \chi_R(Y) |Y|^{-p} |V_{\lambda, c}(Y)|^3 dY - \Theta_0 = \mathcal{F}_0(\lambda, c; V),$$

$$\mathcal{F}_j(1, 0; V_{\lambda, c}) = \int Y_j \chi_R(Y) |V_{\lambda, c}(Y)|^2 dY = \mathcal{F}_j(\lambda, c; V), \quad j = 1, 2, 3.$$

□

Proposition 6.6 (Differentiability of $\mathcal{F}$). For each fixed $V \in \mathcal{A}_{p, R}$, the map $(\lambda, c) \mapsto \mathcal{F}(\lambda, c; V)$ is C^1 in a neighborhood of $(1, 0)$. In each component $\mathcal{F}_\alpha$ all derivatives under the integral sign are legitimate.

Proof. Fix (λ, c) near $(1, 0)$, and abbreviate

$$V_{\lambda, c}(Y) := \lambda^{-1} V(\lambda^{-1}(Y - c)).$$

By direct differentiation,

$$\partial_\lambda V_{\lambda, c} = -\lambda^{-1} V_{\lambda, c} - \lambda^{-1}(Y - c) \cdot \nabla V_{\lambda, c}, \quad \partial_{c_k} V_{\lambda, c} = -\lambda^{-2} \partial_k V(\lambda^{-1}(Y - c)) = -\partial_k V_{\lambda, c}.$$

The neighborhood of $(1, 0)$ is chosen so that the arguments of V remain inside B_{4R} for $Y \in \text{supp} \chi_R$. The scale derivative is controlled by the two weighted integrability conditions

$$\int |Y|^{-p} |V|^3 < \infty, \quad \int |Y|^{-p} |V| |Y \cdot \nabla V| < \infty,$$

transported by the smooth change of variables. The translation derivative in the scale row is treated by the source identity

$$3 \int |Y|^{-p} |V| V \cdot \partial_k V = p \int Y_k |Y|^{-p-2} |V|^3,$$

obtained first for smooth compactly supported profiles and then by approximation in $\mathcal{A}_{p, R}$; the right-hand side is finite because $\int |Y|^{-p-1} |V|^3 < \infty$. The centering rows are supported in B_{2R} and are controlled by $V \in L^2_{\text{loc}}$ and $\nabla V \in L^3_{\text{loc}}$. These bounds give an integrable majorant for the difference quotients, so dominated convergence applies.

$$\mathcal{F}_0(\lambda, c; V) = \int_{\mathbb{R}^3} \chi_R(Y) |Y|^{-p} |V_{\lambda, c}|^3 dY - \Theta_0,$$

for which dominated convergence gives

$$\partial_\lambda \mathcal{F}_0 = \int_{\mathbb{R}^3} \chi_R |Y|^{-p} 3 |V_{\lambda, c}| V_{\lambda, c} \cdot \partial_\lambda V_{\lambda, c} dY,$$

$$\partial_{c_k} \mathcal{F}_0 = \int_{\mathbb{R}^3} \chi_R |Y|^{-p} 3 |V_{\lambda, c}| V_{\lambda, c} \cdot \partial_{c_k} V_{\lambda, c} dY.$$

Similarly

$$\mathcal{F}_j(\lambda, c; V) = \int_{\mathbb{R}^3} Y_j \chi_R(Y) |V_{\lambda, c}|^2 dY, \quad j = 1, 2, 3,$$

hence

$$\partial_\lambda \mathcal{F}_j = \int_{\mathbb{R}^3} 2 Y_j \chi_R V_{\lambda, c} \cdot \partial_\lambda V_{\lambda, c} dY,$$

$$\partial_{c_k} \mathcal{F}_j = \int_{\mathbb{R}^3} 2 Y_j \chi_R V_{\lambda, c} \cdot \partial_{c_k} V_{\lambda, c} dY.$$

Thus each component admits pointwise C^1 formulas and these are continuous in (λ, c) near $(1, 0)$. Therefore $(\lambda, c) \mapsto \mathcal{F}(\lambda, c; V) \in C^1$ near $(1, 0)$. $\square$

Proposition 6.7 (Full local Jacobian at $(1, 0)$). *Let $V \in \mathcal{A}_{p, R}$. Set $V_0(Y) = V(Y)$. At $(\lambda, c) = (1, 0)$, the matrix*

$$J(V) := D_{(\lambda, c)} \mathcal{F}(1, 0; V) \in \mathbb{R}^{4 \times 4}$$

has rows

$$J_\alpha = (\partial_\lambda \mathcal{F}_\alpha, \partial_{c_1} \mathcal{F}_\alpha, \partial_{c_2} \mathcal{F}_\alpha, \partial_{c_3} \mathcal{F}_\alpha), \quad \alpha = 0, \dots, 3.$$

Writing $W_{\text{scl}} = V_0 + Y \cdot \nabla V_0$, one has

$$(1) \quad \partial_\lambda \mathcal{F}_0(1, 0; V) = -3 \int \chi_R |Y|^{-p} |V_0| V_0 \cdot W_{\text{scl}} dY,$$

$$(2) \quad \partial_{c_k} \mathcal{F}_0(1, 0; V) = -3 \int_{\mathbb{R}^3} \chi_R |Y|^{-p} |V_0| V_0 \cdot \partial_k V_0 dY,$$

$$(3) \quad \partial_\lambda \mathcal{F}_j(1, 0; V) = -2 \int_{\mathbb{R}^3} Y_j \chi_R V_0 \cdot W_{\text{scl}} dY,$$

$$(4) \quad \partial_{c_k} \mathcal{F}_j(1, 0; V) = -2 \int_{\mathbb{R}^3} \chi_R Y_j V_0 \cdot \partial_k V_0 dY,$$

for $j, k = 1, 2, 3$.

Proof. Differentiate the formulas in Definition 6.2 at $(\lambda, c) = (1, 0)$.

Since $V_{\lambda, c}(Y) = \lambda^{-1} V(\lambda^{-1}(Y - c))$,

$$\partial_\lambda V_{\lambda, c} = -\lambda^{-1} V_{\lambda, c} - \lambda^{-1} Y \cdot \nabla V_{\lambda, c}, \quad \partial_{c_k} V_{\lambda, c} = -\lambda^{-2} \partial_k V(\lambda^{-1}(Y - c)).$$

At $(1, 0)$, $\partial_\lambda V_{\lambda, c}|_{(1, 0)} = -(V + Y \cdot \nabla V)$, $\partial_{c_k} V_{\lambda, c}|_{(1, 0)} = -\partial_k V$.

For $\mathcal{F}_0$, write

$$\mathcal{F}_0(\lambda, c; V) = \int \chi_R(Y) |Y|^{-p} |V_{\lambda, c}(Y)|^3 dY - \Theta_0.$$

Derivative in λ :

$$\partial_\lambda \mathcal{F}_0 = \int \chi_R |Y|^{-p} 3 |V_{\lambda, c}| V_{\lambda, c} \cdot \partial_\lambda V_{\lambda, c} dY$$

Evaluating at $(1, 0)$ yields (1).

For $k = 1, 2, 3$, c_k -derivative of $\mathcal{F}_0$ has no cutoff term:

$$\partial_{c_k} \mathcal{F}_0(1, 0; V) = \int \chi_R |Y|^{-p} 3 |V| V \cdot (-\partial_k V) dY,$$

which is (2).

For $\mathcal{F}_j$,

$$\mathcal{F}_j(\lambda, c; V) = \int Y_j \chi_R(Y) |V_{\lambda, c}|^2 dY.$$

Hence

$$\partial_\lambda \mathcal{F}_j = \int 2 Y_j \chi_R V_{\lambda, c} \cdot \partial_\lambda V_{\lambda, c} dY,$$

and at $(1, 0)$:

$$\partial_\lambda \mathcal{F}_j(1, 0; V) = -2 \int Y_j \chi_R V \cdot W_{\text{scl}} dY,$$

which is (3). Similarly,

$$\partial_{c_k} \mathcal{F}_j(1, 0; V) = \int 2 Y_j \chi_R V \cdot \partial_{c_k} V_{\lambda, c}|_{(1, 0)} dY = -2 \int \chi_R Y_j V \cdot \partial_k V dY,$$

giving (4). □

Proposition 6.8 (Translation block and Schur-complement invertibility). *Let $V : J \rightarrow \mathcal{A}_{p, R}$ satisfy the repaired gauge constraints*

$$\mathcal{F}(1, 0; V(\tau)) = 0 \quad \text{for a.e. } \tau \in J.$$

Write the modulation matrix in block form

$$J(V(\tau)) = \begin{pmatrix} \alpha(\tau) & u(\tau)^T \\ v(\tau) & A(\tau) \end{pmatrix},$$

where $\alpha = \partial_\lambda \mathcal{F}_0$, $u_\ell = \partial_{c_\ell} \mathcal{F}_0$, $v_j = \partial_\lambda \mathcal{F}_j$, and $A_{j\ell} = \partial_{c_\ell} \mathcal{F}_j$. Assume that for a.e. $\tau \in J$

$$m_\chi(\tau) := \int |V(Y, \tau)|^2 \chi_R(Y) dY \geq m_0 > 0,$$

$$\int_{A_R} |V(Y, \tau)|^2 dY \leq \varepsilon_0, \quad 3C_\chi \varepsilon_0 \leq \frac{m_0}{2},$$

where $A_R = \{R \leq |Y| \leq 2R\}$ and

$$C_\chi := \max_{j,\ell} \|Y_j \partial_\ell \chi_R\|_{L^\infty}.$$

Assume also

$$C_u := \operatorname{ess\,sup}_{\tau \in J} \int |Y|^{-p-1} |V(Y, \tau)|^3 dY < \infty$$

and, with $C_\nabla := \| |Y| \nabla \chi_R \|_{L^\infty}$,

$$6p \left(\frac{2}{m_0} \right) C_u R C_\nabla \varepsilon_0 \leq \frac{p\Theta_0}{2}.$$

Then $J(V(\tau))$ is invertible for a.e. $\tau \in J$, and

$$\operatorname{ess\,sup}_{\tau \in J} \|J(V(\tau))^{-1}\|_{\infty \rightarrow \infty} \leq K(R, p, \Theta_0, m_0, C_u, \varepsilon_0, \chi_R).$$

Proof. The entries are finite by $V(\tau) \in \mathcal{A}_{p,R}$ and the compact support of χ_R . With the present parameter convention, $\partial_\lambda V_{\lambda,c}|_{(1,0)} = -(V + Y \cdot \nabla V)$. Hence the scale entry satisfies the signed transversality identity

$$\alpha(\tau) = -p \int |Y|^{-p} |V(Y, \tau)|^3 dY = -p\Theta_0$$

on the gauge surface. Thus $|\alpha| \geq p\Theta_0$.

For the centering block, integration by parts gives

$$A_{j\ell} = -\delta_{j\ell} \int |V|^2 \chi_R - \int |V|^2 Y_j \partial_\ell \chi_R.$$

Writing $A = -m_\chi I_3 + E$, the annular support of $\nabla \chi_R$ gives

$$|E_{j\ell}| \leq C_\chi \int_{A_R} |V|^2 \leq C_\chi \varepsilon_0, \quad \|E\|_{\infty \rightarrow \infty} \leq 3C_\chi \varepsilon_0 \leq m_0/2.$$

Thus $A = -m_\chi(I_3 - m_\chi^{-1}E)$ and $\|m_\chi^{-1}E\|_{\infty \rightarrow \infty} \leq 1/2$, so A^{-1} exists and $\|A^{-1}\|_{\infty \rightarrow \infty} \leq 2/m_0$.

For the scale-to-translation block, the repaired scale-row formula gives

$$|u_\ell| \leq p \int |Y|^{-p-1} |V|^3 dY \leq p \int |Y|^{-p-1} |V|^3 dY,$$

so $\|u\|_\infty \leq pC_u$. For the translation-to-scale block,

$$v_j = - \int |V|^2 Y_j (Y \cdot \nabla \chi_R) dY$$

because the centering constraint cancels the interior term. Therefore

$$\|v\|_\infty \leq 2R C_\nabla \varepsilon_0.$$

The Schur complement $S = \alpha - u^T A^{-1} v$ satisfies

$$|u^T A^{-1} v| \leq 3\|u\|_\infty \|A^{-1}\|_{\infty \rightarrow \infty} \|v\|_\infty \leq 6p \left(\frac{2}{m_0} \right) C_u R C_\nabla \varepsilon_0 \leq \frac{p\Theta_0}{2}.$$

Since $|\alpha| = p\Theta_0$, this gives $|S| \geq p\Theta_0/2$. The block inverse formula

$$J^{-1} = \begin{pmatrix} S^{-1} & -S^{-1}u^T A^{-1} \\ -A^{-1}vS^{-1} & A^{-1} + A^{-1}vS^{-1}u^T A^{-1} \end{pmatrix}$$

then gives the displayed bound. $\square$

7. REPAIRED GAUGE BY IMPLICIT INVERSION IN TIME

Lemma 7.1 (Time chain rule for preliminary profile). *For each $\alpha = 0, \dots, 3$, set $G_\alpha(\tau) := \mathcal{F}_\alpha(1, 0; V_{\text{pre}}(\tau))$. Then $G_\alpha \in W_{\text{loc}}^{1,1}(I_*)$ and for a.e. τ ,*

$$G'_\alpha(\tau) = D_V \mathcal{F}_\alpha(1, 0; V_{\text{pre}}(\tau)) [\partial_\tau V_{\text{pre}}(\tau)].$$

The right-hand side is obtained from the profile formulas of Proposition 6.6 and is in $L^1_{\text{loc}}(I_)$.*

Proof. Let $V_{\text{pre}}^\varepsilon$ be a spatial and temporal Steklov mollification inside a slightly larger compact cylinder. For the smoothed profiles the ordinary chain rule gives

$$\frac{d}{d\tau} \mathcal{F}_\alpha(1, 0; V_{\text{pre}}^\varepsilon(\tau)) = D_V \mathcal{F}_\alpha(1, 0; V_{\text{pre}}^\varepsilon(\tau)) [\partial_\tau V_{\text{pre}}^\varepsilon(\tau)].$$

The transformed equation expresses $\partial_\tau V_{\text{pre}}^\varepsilon$ as the mollification of $\nu \Delta V_{\text{pre}} - (V_{\text{pre}} \cdot \nabla) V_{\text{pre}} - \nabla P_{\text{pre}}$ plus the chart transport terms. Pairing this distribution with the compactly supported Gateaux gradients below is legitimate by the local suitable bounds, the pressure bound, and the weighted L^3 control in (SC4). Passing $\varepsilon \downarrow 0$ gives the asserted $W_{\text{loc}}^{1,1}$ identity. For $W \in \mathcal{V}_R$, these derivative formulas hold:

$$\begin{aligned} D_V \mathcal{F}_0(1, 0; V)[W] &= \int_{\mathbb{R}^3} 3\chi_R |Y|^{-p} |V| V \cdot W \, dY, \\ D_V \mathcal{F}_j(1, 0; V)[W] &= \int_{\mathbb{R}^3} 2Y_j \chi_R V \cdot W \, dY, \quad j = 1, 2, 3, \end{aligned}$$

where both right-hand sides are finite on $\mathcal{V}_R$ by compact support, $p < 3$, and $V, W \in L^3 \cap W_{\text{loc}}^{1,3/2}$. The right-hand side is finite by the explicit formulas above and by the distributional equation paired with compactly supported test gradients; the local energy, pressure, and weighted L^3 bounds give an $L^1_{\text{loc}}(I_*)$ majorant. $\square$

Lemma 7.2 (AC implicit inversion with parameters). *Let $U \subset \mathbb{R}^4$ be open and $G : I_* \times U \rightarrow \mathbb{R}^4$. Assume:*

- (i) *for each $s \in U$, $\tau \mapsto G(\tau, s) \in W_{\text{loc}}^{1,1}(I_*; \mathbb{R}^4)$, and $D_\tau G \in L^1_{\text{loc}}(I_*; L^\infty(U; \mathbb{R}^4))$;*
- (ii) *for a.e. $\tau \in I_*$, $s \mapsto G(\tau, s)$ is C^1 on U , with $D_s G$ continuous in (τ, s) ;*
- (iii) *for some $(\tau_0, s_0) \in I_* \times U$, $G(\tau_0, s_0) = 0$, and there is $\rho, C > 0$ such that*

$$\sup_{(\tau, s) \in I_* \times B_\rho(s_0)} \|D_s G(\tau, s)^{-1}\|_{\mathcal{L}(\mathbb{R}^4, \mathbb{R}^4)} \leq C.$$

Then after possibly shrinking I_ to a subinterval $J \ni \tau_0$, there is a unique map $s \in AC(J; U)$, $s(\tau_0) = s_0$, with*

$$G(\tau, s(\tau)) = 0 \quad \text{a.e. in } J.$$

Moreover

$$s'(\tau) = -D_s G(\tau, s(\tau))^{-1} D_\tau G(\tau, s(\tau)) \quad \text{a.e. on } J.$$

Proof. By (ii), the matrix field $D_s G$ is continuous in s , so there is $r_0 \in (0, \rho)$ such that $\inf_{(\tau, s) \in I_* \times B_{r_0}(s_0)} |\det D_s G(\tau, s)| > 0$ after possibly shrinking I_* ; this keeps $D_s G^{-1}$ bounded and continuous on $B_{r_0}(s_0)$.

Set

$$F(\tau, s) := -D_s G(\tau, s)^{-1} D_\tau G(\tau, s).$$

Then $F(\tau, \cdot)$ is continuous on $B_{r_0}(s_0)$ for a.e. τ and $F(\cdot, s) \in L^1_{\text{loc}}(I_*; \mathbb{R}^4)$ by (i), hence F is a Carathéodory right-hand side. Since the gauge map is C^1 in the finite variables, after shrinking r_0 there is $L \in L^1(J)$ such that

$$|F(\tau, s_1) - F(\tau, s_2)| \leq L(\tau) |s_1 - s_2| \quad \text{for } s_1, s_2 \in B_{r_0}(s_0).$$

Choose $\delta > 0$ so small that

$$\int_{\tau_0 - \delta}^{\tau_0 + \delta} \sup_{s \in B_{r_0}(s_0)} |F(\sigma, s)| d\sigma < r_0.$$

On $J = [\tau_0 - \delta, \tau_0 + \delta] \cap I_*$ define the Picard operator

$$(\mathcal{T}s)(\tau) := s_0 + \int_{\tau_0}^{\tau} F(\sigma, s(\sigma)) d\sigma$$

on $C(J; B_{r_0}(s_0))$. The preceding bound keeps $\mathcal{T}$ in the same closed ball, and the Lipschitz estimate gives

$$\|\mathcal{T}s_1 - \mathcal{T}s_2\|_{C(J)} \leq \left(\int_J L(\sigma) d\sigma \right) \|s_1 - s_2\|_{C(J)}.$$

After decreasing δ once more, the coefficient is < 1 , so Banach's fixed-point theorem gives a unique solution

$$s(\tau) = s_0 + \int_{\tau_0}^{\tau} F(\sigma, s(\sigma)) d\sigma, \quad \tau \in J,$$

which stays in $B_{r_0}(s_0) \subset U$. This solution is absolutely continuous.

Define $H(\tau) := G(\tau, s(\tau))$. Since $G(\cdot, s(\cdot)) \in W_{\text{loc}}^{1,1}$ from (i), the chain rule gives for a.e. $\tau \in J$

$$H'(\tau) = D_\tau G(\tau, s(\tau)) + D_s G(\tau, s(\tau))s'(\tau) = D_\tau G(\tau, s(\tau)) + D_s G(\tau, s(\tau))F(\tau, s(\tau)) = 0.$$

Hence $H \equiv G(\tau_0, s_0) = 0$ on J , which is the claimed constraint. Uniqueness follows from Grönwall on the difference of two AC solutions on $B_{r_0}(s_0)$, using Lipschitz continuity of $F(\tau, \cdot)$ on that ball. $\square$

Definition 7.3 (Retained nondegenerate repaired-gauge stratum). Let $J \subset I_*$ be compact. The retained nondegenerate repaired-gauge stratum over the preliminary profile V_{pre} consists of branches for which there are

$$\lambda \in AC(J; (0, \infty)), \quad c \in AC(J; \mathbb{R}^3),$$

such that, for a.e. $\tau \in J$,

$$V(\tau) := V_{\text{pre}}^{\lambda(\tau), c(\tau)} \in \mathcal{A}_{p,R}, \quad \mathcal{F}(1, 0; V(\tau)) = 0.$$

The path is quantitatively nondegenerate if the translation-block, annular, weighted first-moment, and Schur-complement bounds in Proposition 6.8 hold on J . If no such local AC selection exists, or if the quantitative nondegeneracy package fails persistently, the branch is not in this stratum; it is routed to the gauge-degenerate, multibubble, or cascade alternatives.

Proposition 7.4 (Gauge regularity on the retained stratum). *Let (λ, c) be the repaired-gauge selection of the retained nondegenerate stratum on J . Then*

$$\mathcal{F}(\lambda(\tau), c(\tau); V_{\text{pre}}(\tau)) = 0 \quad \text{for a.e. } \tau \in J,$$

and $(\lambda, c) \in AC(J; \mathbb{R}^4)$ by definition. If the path is quantitatively nondegenerate, then

$$\text{ess sup}_{\tau \in J} \|J(V(\tau))^{-1}\| < \infty.$$

Proof. The constraint follows from Lemma 6.5 applied to $V(\tau) = V_{\text{pre}}^{\lambda(\tau), c(\tau)}$. The absolute continuity is part of Definition 7.3. The final assertion is exactly Proposition 6.8. $\square$

Definition 7.5 (Repaired chart). Define

$$\begin{aligned} \Lambda(\tau) &:= \Lambda_{\text{pre}}(\tau)\lambda(\tau), & X(\tau) &:= X_{\text{pre}}(\tau) + \Lambda_{\text{pre}}(\tau)c(\tau), \\ V(\tau) &:= V_{\text{pre}}^{\lambda(\tau), c(\tau)}, & P(\tau) &:= P_{\text{pre}}^{\lambda(\tau), c(\tau)} \text{ (same physical pressure pull-back under } \Phi). \end{aligned}$$

Theorem 7.6 (AC repaired parameters). *Assume $\text{SC}(R_*, r_*, M, \eta, \kappa)$, and suppose the branch lies in the retained nondegenerate repaired-gauge stratum on J , with selection (λ, c) . Then the corrected chart (Λ, X) is absolutely continuous on J , satisfies*

$$a(\tau) = -\Lambda'(\tau) \in L^1_{\text{loc}}(J), \quad b(\tau) = -X'(\tau) \in L^1_{\text{loc}}(J),$$

and the gauge constraints $\mathcal{F}(1, 0; V(\tau)) = 0$ hold a.e. Equivalently,

$$\mathcal{F}(\lambda(\tau), c(\tau); V_{\text{pre}}(\tau)) = 0$$

for the repaired-gauge selection (λ, c) .

Proof. By Definition 7.3, $(\lambda, c) \in \text{AC}(J)$ and the repaired gauge constraints hold for $V(\tau)$ a.e. By definition $V(\tau) = V_{\text{pre}}^{\lambda(\tau), c(\tau)}$, so Lemma 6.5 applied at $(\lambda, c) = (\lambda(\tau), c(\tau))$ gives

$$\mathcal{F}(1, 0; V_{\text{pre}}^{\lambda, c}) = \mathcal{F}(\lambda, c; V_{\text{pre}})$$

implies $\mathcal{F}(1, 0; V(\tau)) = 0$ a.e.

Since $\Lambda = \Lambda_{\text{pre}}\lambda$ and $X = X_{\text{pre}} + \Lambda_{\text{pre}}c$, and $\Lambda_{\text{pre}}, X_{\text{pre}}$ are AC by (SC5), the products and sums preserve AC, hence $(\Lambda, X) \in \text{AC}(I_*)$. Differentiating,

$$a = -\Lambda' = -(\Lambda'_{\text{pre}}\lambda + \Lambda_{\text{pre}}\lambda') \in L^1_{\text{loc}}, \quad b = -X' = -(X'_{\text{pre}} + \Lambda'_{\text{pre}}c + \Lambda_{\text{pre}}c') \in L^1_{\text{loc}}.$$

This proves the claimed regularity. $\square$

Remark 7.7 (Gauge degeneracy). Failure of invertibility of J is the gauge-degenerate exit from the retained single-core stratum for this fixed renormalization mechanism. In the Type II state-space stratification this exit is handled with the multibubble/cascade and scale-rigidity alternatives, not inside the present representation theorem.

8. PRESSURE TRANSPORT AND DECOMPOSITION

Lemma 8.1 (Renormalized pressure equation). *If V, P satisfy the renormalized Navier–Stokes equation in Theorem 3.2, then*

$$-\Delta P = \partial_i \partial_j (V_i V_j) \quad \text{in } \mathcal{D}'(Q_{2R}^*).$$

Proof. Take test $\varphi \in C_c^\infty(Q_{2R}^*)$. Since $V, P \in L^2_{\text{loc}} \times L^{3/2}_{\text{loc}}$ on Q_{2R}^* , all pairings below are legitimate. Expand

$$\nabla \cdot ((V \cdot \nabla)V) = \sum_{i,j} \partial_i \partial_j (V_i V_j) - \sum_{i,j} \partial_i V_i \partial_j V_j.$$

Because $\nabla \cdot V = 0$, this reduces to $\partial_i \partial_j (V_i V_j)$. Taking divergence of

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P - \nu \Delta V + a(V + Y \cdot \nabla V) + b \cdot \nabla V = 0$$

and using $\nabla \cdot V = 0$, $\nabla \cdot (V + Y \cdot \nabla V) = 0$, $\nabla \cdot (b \cdot \nabla V) = b \cdot \nabla (\nabla \cdot V) = 0$, and $\nabla \cdot \Delta V = \Delta (\nabla \cdot V) = 0$, we get in $\mathcal{D}'$:

$$\Delta P + \partial_i \partial_j (V_i V_j) = 0.$$

Hence $-\Delta P = \partial_i \partial_j (V_i V_j)$ on Q_{2R}^* . $\square$

Theorem 8.2 (Localized pressure decomposition). *Let $\zeta \in C_c^\infty(B_{2R})$, $\zeta \equiv 1$ on B_R , and assume $V \in L^1_{\text{loc}}(I_*; L^3(B_{2R}))$, $P \in L^1_{\text{loc}}(I_*; L^{3/2}(B_{2R}))$. For a.e. $\tau \in I_*$ there exists*

$$P(\cdot, \tau) = P_{\text{loc}}(\cdot, \tau) + H(\cdot, \tau)$$

on B_R such that

$$\begin{cases} -\Delta P_{\text{loc}}(\cdot, \tau) = \partial_i \partial_j (\zeta V_i V_j), \\ -\Delta H(\cdot, \tau) = 0 \text{ on } B_R, \quad \int_{B_R} H(\cdot, \tau) dY = 0. \end{cases}$$

Moreover

$$\|P_{\text{loc}}(\tau)\|_{L^{3/2}(B_R)} \lesssim \|V(\tau)\|_{L^3(B_{2R})}^2, \quad \|H(\tau)\|_{L^q(B_r)} \lesssim_{q,R,r} \|P(\tau)\|_{L^{3/2}(B_{2R})} + \|V(\tau)\|_{L^3(B_{2R})}^2$$

for each $1 \leq q < \infty$ and each $0 < r < R$, with constants depending only on R , r , and q .

Proof. For fixed τ , set

$$P_{\text{loc}} = \mathcal{R}_i \mathcal{R}_j (\zeta V_i V_j), \quad H := P - P_{\text{loc}}.$$

$\zeta V_i V_j \in L_{\text{loc}}^1(I_*; L^{3/2}(B_{2R}))$, so the Calderón–Zygmund operator $f \mapsto \mathcal{R}_i \mathcal{R}_j f$ gives

$$P_{\text{loc}}(\cdot, \tau) = \mathcal{R}_i \mathcal{R}_j (\zeta V_i V_j)(\cdot, \tau), \quad \|P_{\text{loc}}(\tau)\|_{L^{3/2}(B_R)} \lesssim \|V(\tau)\|_{L^3(B_{2R})}^2.$$

Because $\mathcal{R}_i \mathcal{R}_j$ is linear bounded on $L^{3/2}$,

$$P_{\text{loc}} \in L_{\text{loc}}^1(I_*; L^{3/2}(B_R)).$$

Since $-\Delta P_{\text{loc}} = \partial_i \partial_j (\zeta V_i V_j)$ and $\zeta \equiv 1$ on B_R ,

$$-\Delta H(\cdot, \tau) = \partial_i \partial_j ((1 - \zeta) V_i V_j) = 0 \quad \text{in } B_R,$$

so $H(\cdot, \tau)$ is harmonic on B_R . Imposing $\oint_{B_R} H = 0$ fixes the time-dependent additive constant uniquely in each time slice. For $q \in [1, \infty)$ and $0 < r < R$, harmonicity and local L^p -to- L^q interior estimates imply

$$\|H(\tau)\|_{L^q(B_r)} \lesssim_{q,R,r} \|H(\tau)\|_{L^{3/2}(B_R)}.$$

Finally, $\|H(\tau)\|_{L^{3/2}(B_R)} \leq \|P(\tau)\|_{L^{3/2}(B_R)} + \|P_{\text{loc}}(\tau)\|_{L^{3/2}(B_R)}$, giving the announced bound in terms of P and V . $\square$

Remark 8.3 (Measurability and normalization). If V, P are measurable in τ , then $\tau \mapsto P_{\text{loc}}(\cdot, \tau)$ is measurable in $L^{3/2}(B_R)$ by boundedness of $\mathcal{R}_i \mathcal{R}_j$, hence $\tau \mapsto H(\cdot, \tau)$ is measurable in each $L^q(B_r)$, $0 < r < R$. The normalization $\oint_{B_R} H(\cdot, \tau) dY = 0$ fixes the time-dependent additive constant in H uniquely for each τ , and therefore the decomposition is a legitimate renormalized pressure splitting.

9. MODULATION FROM DIFFERENTIATED CONSTRAINTS

Lemma 9.1 (Differentiation of constraints). *For the repaired profile V constructed above, with $|Y|^{-p}|V|^3 \in L_{\text{loc}}^1(I_*; L^1)$, Then $G_\alpha(\tau) := \mathcal{F}_\alpha(1, 0; V(\tau))$ are absolutely continuous and*

$$\frac{d}{d\tau} G_\alpha(\tau) = \int_{\mathbb{R}^3} \mathbf{D}_\alpha(V, \nabla V, \nabla \chi_R) \partial_\tau V dY$$

for a.e. τ . Here $\mathbf{D}_\alpha$ is the L^1 -integrable Gateaux gradient associated to $\mathcal{F}_\alpha$.

Proof.

$$G_\alpha(\tau) = \mathcal{F}_\alpha(1, 0; V(\tau))$$

is first evaluated on spatial-temporal mollifications V^ε . For those smooth profiles the ordinary chain rule gives

$$\frac{d}{d\tau} G_\alpha^\varepsilon(\tau) = D_V \mathcal{F}_\alpha(1, 0; V^\varepsilon(\tau)) [\partial_\tau V^\varepsilon(\tau)].$$

The local suitable bounds, pressure estimates, and compact support of the Gateaux gradients below allow passage to the limit $\varepsilon \downarrow 0$, which gives the displayed derivative formula for V . For $\mathcal{F}_0$,

$$\mathbf{D}_0(V, \nabla V, \nabla \chi_R) = 3\chi_R |Y|^{-p} |V| V;$$

for $\alpha = 1, 2, 3$,

$$\mathbf{D}_\alpha(V, \nabla V, \nabla \chi_R) = 2\chi_R Y_\alpha V.$$

Each coefficient is $L^1(B_{2R})$: $3\chi_R|Y|^{-p}|V|V \in L^{3/2}(B_{2R}) \subset L^1(B_{2R})$ from $p < 3$ and $|Y|^{-p}|V|^3 \in L^1$, and $2\chi_R Y_\alpha V \in L^2(B_{2R}) \subset L^1(B_{2R})$ by compact support.

$$\frac{d}{d\tau} G_\alpha(\tau) = \int_{\mathbb{R}^3} \mathbf{D}_\alpha(V, \nabla V, \nabla \chi_R) \cdot \partial_\tau V \, dY,$$

Since $\mathbf{D}_\alpha(V, \nabla V, \nabla \chi_R) \in L^1(B_{2R})$ and $\partial_\tau V \in L^1_{\text{loc}}(I_*; L^2(B_{2R}))$, the right-hand side is in $L^1_{\text{loc}}(I_*)$, so $G_\alpha \in W^{1,1}_{\text{loc}}(I_*) \subset AC_{\text{loc}}(I_*)$. $\square$

Lemma 9.2 (Directional derivatives in the profile variable). *For $\mathcal{A}_\tau = V + Y \cdot \nabla V$ and $\alpha = 0, 1, 2, 3$,*

$$D_V \mathcal{F}_\alpha(1, 0; V)[\mathcal{A}_\tau] = -\partial_\lambda \mathcal{F}_\alpha(1, 0; V), \quad D_V \mathcal{F}_\alpha(1, 0; V)[\partial_k V] = -\partial_{c_k} \mathcal{F}_\alpha(1, 0; V).$$

Indeed, from $V_{\lambda,c}(Y) = \lambda^{-1}V(\lambda^{-1}(Y - c))$,

$$\partial_\lambda V_{\lambda,c}|_{(1,0)} = -(V + Y \cdot \nabla V) = -\mathcal{A}_\tau, \quad \partial_{c_k} V_{\lambda,c}|_{(1,0)} = -\partial_k V.$$

Substituting these into the Gateaux formulas from Proposition 6.6 gives the two identities.

Theorem 9.3 (Modulation system). *Let (V, P) be the repaired pair on the compact window under consideration. Writing $\theta := (a, b)$, define*

$$\mathcal{A}_\tau(Y) := V(\tau, Y) + Y \cdot \nabla V(\tau, Y).$$

For $\alpha = 0, 1, 2, 3$, define

$$\mathcal{R}_\alpha(\tau) := - \int_{\mathbb{R}^3} \mathbf{D}_\alpha(V, \nabla V, \nabla \chi_R) \cdot ((V \cdot \nabla)V + \nabla P - \nu \Delta V) \, dY.$$

Then for a.e. $\tau \in I_*$,

$$\partial_\lambda \mathcal{F}_\alpha(1, 0; V) a(\tau) + \sum_{k=1}^3 \partial_{c_k} \mathcal{F}_\alpha(1, 0; V) b_k(\tau) = \mathcal{R}_\alpha(\tau),$$

Equivalently,

$$J(\tau)\theta(\tau) = \mathcal{R}(\tau).$$

Hence

$$\theta(\tau) = J(\tau)^{-1} \mathcal{R}(\tau) \in L^1_{\text{loc}}(I_*)^4, \quad |\theta(\tau)| \leq \kappa |\mathcal{R}(\tau)|.$$

Proof. Set $G_\alpha(\tau) := \mathcal{F}_\alpha(1, 0; V(\tau))$. By Theorem 7.6, $G_\alpha(\tau) = 0$ a.e.; by Lemma 9.1, $G_\alpha \in W^{1,1}_{\text{loc}}$. Hence for a.e. τ ,

$$0 = \frac{d}{d\tau} G_\alpha(\tau) = D_V \mathcal{F}_\alpha(1, 0; V(\tau)) [\partial_\tau V(\tau)].$$

Insert the equation from Theorem 3.2,

$$\partial_\tau V = -(V \cdot \nabla)V - \nabla P + \nu \Delta V - a \mathcal{A}_\tau - b \cdot \nabla V.$$

This gives

$$0 = -D_V \mathcal{F}_\alpha[(V \cdot \nabla)V + \nabla P - \nu \Delta V] - a D_V \mathcal{F}_\alpha[\mathcal{A}_\tau] - \sum_{k=1}^3 b_k D_V \mathcal{F}_\alpha[\partial_k V].$$

By Lemma 9.1, for any profile direction W ,

$$D_V \mathcal{F}_\alpha(1, 0; V)[W] = \int_{\mathbb{R}^3} \mathbf{D}_\alpha(V, \nabla V, \nabla \chi_R) \cdot W \, dY.$$

By Lemma 9.2,

$$D_V \mathcal{F}_\alpha[\mathcal{A}_\tau] = -\partial_\lambda \mathcal{F}_\alpha(1, 0; V), \quad D_V \mathcal{F}_\alpha[\partial_k V] = -\partial_{c_k} \mathcal{F}_\alpha(1, 0; V).$$

Substituting into the previous line yields the linear system

$$\partial_\lambda \mathcal{F}_\alpha(1, 0; V) a(\tau) + \sum_{k=1}^3 \partial_{c_k} \mathcal{F}_\alpha(1, 0; V) b_k(\tau) = \mathcal{R}_\alpha(\tau),$$

which is equivalent to $J(\tau)\theta(\tau) = \mathcal{R}(\tau)$. By definition of $\mathbf{D}_\alpha$, each $\mathcal{R}_\alpha \in L^1_{\text{loc}}(I_*)$, and Proposition 6.8 gives $|J^{-1}(\tau)| \leq K$ a.e., hence

$$\theta(\tau) = J^{-1}(\tau)\mathcal{R}(\tau), \quad \theta \in L^1_{\text{loc}}(I_*), \quad |\theta| \leq K|\mathcal{R}| \text{ a.e..}$$

□

Corollary 9.4 (Regularity of a, b). *For the repaired pair of Theorem 9.3, $a, b \in L^1_{\text{loc}}(I_*)$. If in addition $\mathcal{R} \in L^\infty_{\text{loc}}(I_*)$, then $a, b \in L^\infty_{\text{loc}}(I_*)$.*

Proof. By Theorem 9.3, $\theta = (a, b) = J^{-1}\mathcal{R}$ a.e., and $J^{-1} \in L^\infty_{\text{loc}}(I_*)$ by Theorem 6.8. The first claim is therefore immediate from $\mathcal{R} \in L^\infty_{\text{loc}}$, while the second follows if $\mathcal{R} \in L^\infty_{\text{loc}}$. □

10. REPAIRED SUITABLE STRUCTURE IN RENORMALIZED VARIABLES

Proposition 10.1 (Transport of suitable local energy inequality). *Suppose (u, p) is suitable on $\mathcal{Q}^{\text{ph}}_{2R}$, and (V, P) is obtained by the repaired map Φ . Set*

$$\tilde{a}(\tau) := -\Lambda(\tau)^{-1}\Lambda'(\tau), \quad \tilde{b}(\tau) := -\Lambda(\tau)^{-1}X'(\tau).$$

Then for every nonnegative $\phi \in C_c^\infty(Q_R^)$ and $\tau_1 < \tau_2$,*

$$\begin{aligned} \mathcal{E}_\phi(V, P) &:= \frac{1}{2}|V|^2\partial_\tau(\phi^2) + \frac{\nu}{2}|V|^2\Delta_Y\phi^2 + \frac{1}{2}|V|^2V \cdot \nabla(\phi^2) \\ &\quad + (P_{\text{loc}} + H)V \cdot \nabla(\phi^2) + \frac{\tilde{a}}{2}|V|^2\phi^2 + \frac{\tilde{a}}{2}|V|^2Y \cdot \nabla(\phi^2) + \frac{1}{2}|V|^2\tilde{b} \cdot \nabla(\phi^2). \end{aligned}$$

$$\begin{aligned} \frac{1}{2} \int_{B_R} |V(\tau_2)|^2 \phi(\cdot, \tau_2)^2 + \nu \int_{\tau_1}^{\tau_2} \int_{B_R} |\nabla V|^2 \phi^2 dY d\tau &\leq \frac{1}{2} \int_{B_R} |V(\tau_1)|^2 \phi(\cdot, \tau_1)^2 \\ &\quad + \int_{\tau_1}^{\tau_2} \int_{B_R} \mathcal{E}_\phi(V, P) dY d\tau. \end{aligned}$$

The right-hand side is controlled on compact windows by $\|\tilde{a}\|_{L^1}, \|\tilde{b}\|_{L^1}, M$, and the local bounds from Theorem 8.2, and support bounds from Section 4. Hence (V, P) is locally suitable on Q_R^ .*

Proof. Apply the physical local energy inequality on $\mathcal{Q}^{\text{ph}}_{2R}$: for nonnegative $\psi \in C_c^\infty(\mathcal{Q}^{\text{ph}}_{2R})$,

$$\frac{1}{2} \int |u|^2 \psi|_{t_1}^{t_2} + \nu \int_{t_1}^{t_2} \int |\nabla_x u|^2 \psi dx dt \leq \int_{t_1}^{t_2} \int \left(\frac{1}{2}|u|^2(\partial_t \psi + \nu \Delta_x \psi) + \left(\frac{|u|^2}{2}u + p u \right) \cdot \nabla_x \psi \right) dx dt.$$

Choose

$$\psi(x, t) := \Lambda(t)^{-1} \phi\left(\frac{x - X(t)}{\Lambda(t)}, \tau(t)\right)^2, \quad \tau = \tau(t), \quad d\tau/dt = \Lambda^{-2},$$

let $t_i = t(\tau_i)$, $i = 1, 2$. By Proposition 2.1 and Section 4, $\psi \in C_c^\infty(\mathcal{Q}^{\text{ph}}_{2R})$.

$$\nabla_x \psi = \Lambda^{-2} \nabla_Y \phi^2, \quad \Delta_x \psi = \Lambda^{-3} \Delta_Y \phi^2, \quad \partial_t \psi = -\Lambda^{-4} \Lambda' \phi^2 + \Lambda^{-3} \partial_\tau(\phi^2) - \Lambda^{-4} (X' + \Lambda' Y) \cdot \nabla_Y \phi^2.$$

Using $u = \Lambda^{-1}V$, $p = \Lambda^{-2}P$, $\nabla_x u = \Lambda^{-2} \nabla_Y V$, $dx dt = \Lambda^5 dY d\tau$, each term in the physical inequality becomes

$$\frac{1}{2} \int |u|^2 \psi|_{t_1}^{t_2} = \frac{1}{2} \int |V|^2 \phi^2|_{\tau_1}^{\tau_2},$$

$$\begin{aligned}
\nu \int |\nabla_x u|^2 \psi \, dx dt &= \nu \int |\nabla V|^2 \phi^2 \, dY \, d\tau, \\
\frac{1}{2} \int |u|^2 \partial_t \psi \, dx dt &= \frac{1}{2} \int |V|^2 \left(\partial_\tau(\phi^2) + \tilde{a} \phi^2 + \tilde{a} Y \cdot \nabla_Y \phi^2 + \tilde{b} \cdot \nabla_Y \phi^2 \right) \, dY \, d\tau, \\
\frac{\nu}{2} \int |u|^2 \Delta_x \psi \, dx dt &= \frac{\nu}{2} \int |V|^2 \Delta_Y \phi^2 \, dY \, d\tau, \\
\int \left(\frac{|u|^2}{2} u + p u \right) \cdot \nabla_x \psi \, dx dt &= \int \left(\frac{1}{2} |V|^2 V + (P_{\text{loc}} + H) V \right) \cdot \nabla_Y \phi^2 \, dY \, d\tau.
\end{aligned}$$

Substituting these identities and grouping terms gives the proposition's right-hand side and coefficients $\tilde{a}, \tilde{b}$. $\square$

11. STABILITY UNDER SUBSEQUENCES

Proposition 11.1. *Let $(u^{(n)}, p^{(n)})$ satisfy $\text{SC}(R_*, r_*, M, \eta, \kappa)$ with the same data parameters and let $(\Lambda^{(n)}, X^{(n)}, V^{(n)}, P^{(n)}, a^{(n)}, b^{(n)})$ be obtained from retained quantitatively nondegenerate repaired-gauge selections on the same I_* , with the constants in Proposition 6.8 uniform in n . After extraction $n_j \rightarrow \infty$, there exist subsequences and limits (Λ, X, V, P, a, b) such that*

$$\begin{aligned}
\Lambda^{(n_j)} &\rightarrow \Lambda \text{ in } L_{\text{loc}}^1, & X^{(n_j)} &\rightarrow X \text{ in } L_{\text{loc}}^1, \\
\Lambda^{(n_j)'} &\rightharpoonup \Lambda' \text{ in } L_{\text{loc}}^1, & X^{(n_j)'} &\rightharpoonup X' \text{ in } L_{\text{loc}}^1, \\
V^{(n_j)} &\rightarrow V \text{ in } L_{\text{loc}}^2(I_*; L_{\text{loc}}^2), & V^{(n_j)} &\rightharpoonup V \text{ in } L_{\text{loc}}^2(I_*; H_{\text{loc}}^1), \\
P^{(n_j)} &\rightharpoonup P \text{ in } L_{\text{loc}}^{3/2}(Q_R^*), & (a^{(n_j)}, b^{(n_j)}) &\rightharpoonup (a, b) \text{ in } L_{\text{loc}}^1(I_*)^4.
\end{aligned}$$

Moreover $a = -\Lambda'$, $b = -X'$ a.e. on I_* , and the limit obeys all conclusions below.

Proof. From SC and the retained repaired-gauge selections, $(V^{(n)}, P^{(n)})$ are uniformly bounded in $L_\tau^\infty L^2(B_{2R}) \cap L_\tau^2 H^1(B_{2R})$ and $L_{\text{loc}}^{3/2}(Q_R^*)$. Theorem 6.8 gives uniform invertibility bounds for $J^{(n)}$, and Theorem 9.3 gives $(a^{(n)}, b^{(n)})$ uniformly in L_{loc}^1 (since $\mathcal{R}^{(n)} \in L_{\text{loc}}^1$). Hence $\Lambda^{(n)}$ and $X^{(n)}$ are uniformly bounded in $W_{\text{loc}}^{1,1}$, so by BV compactness and 1D subsequences,

$$\Lambda^{(n_j)} \rightarrow \Lambda \text{ in } L_{\text{loc}}^1, \quad X^{(n_j)} \rightarrow X \text{ in } L_{\text{loc}}^1,$$

and $a^{(n_j)} \rightharpoonup -\Lambda'$, $b^{(n_j)} \rightharpoonup -X'$ in L_{loc}^1 . From the renormalized equation,

$$\partial_\tau V^{(n)} \in L_\tau^1(H_{\text{loc}}^{-1}) + L_\tau^{4/3}(L_{\text{loc}}^{12/11}),$$

uniformly on compact windows. Aubin–Lions compactness on bounded balls gives

$$V^{(n_j)} \rightarrow V \text{ in } L_{\text{loc}}^2(I_*; L_{\text{loc}}^2), \quad V^{(n_j)} \rightharpoonup V \text{ in } L_{\text{loc}}^2(I_*; H_{\text{loc}}^1).$$

By Calderón–Zygmund and Theorem 8.2, $P^{(n_j)} \rightharpoonup P$ in $L_{\text{loc}}^{3/2}(Q_R^*)$, and with normalized pressure split $P = P_{\text{loc}} + H$, the same for P_{loc} and H . Passing to limits in $\mathcal{F}(1, 0; V^{(n)}(\tau)) = 0$, Theorem 3.2, and the linear modulation system yields the corresponding limit constraints and equations. $\square$

12. FINAL REPRESENTATION PACKAGE

Theorem 12.1 (Chart and repaired-gauge representation). *Assume $\text{SC}(R_*, r_*, M, \eta, \kappa)$, and suppose the branch lies in the retained repaired-gauge stratum with selection (λ, c) on $J = [\tau_0 - \ell, \tau_0 + \ell] \subset I_*$. Then there are fields (V, P) on Q_{2R}^J , with $(\Lambda, X) \in AC(J)$, $a = -\Lambda'$, $b = -X' \in L_{\text{loc}}^1(J)$, such that*

$$u(x, t) = \Lambda(\tau)^{-1} V\left(\frac{x - X(\tau)}{\Lambda(\tau)}, \tau\right), \quad \frac{dt}{d\tau} = \Lambda(\tau)^2,$$

and $\mathcal{F}(1, 0; V(\tau)) = 0$ for $\tau \in J$.

Proof. From (SC5) and Proposition 2.1, $(\Lambda_{\text{pre}}, X_{\text{pre}})$ are AC on I_* , so V_{pre} is defined on Q_{2R}^* . The retained repaired-gauge stratum supplies $(\lambda, c) \in AC(J; (0, \infty) \times \mathbb{R}^3)$ and $\mathcal{F}(1, 0; V_{\text{pre}}^{\lambda, c}) = 0$ a.e. By Definition 7.5 and Theorem 7.6, setting

$$\Lambda(\tau) = \Lambda_{\text{pre}}(\tau)\lambda(\tau), \quad X(\tau) = X_{\text{pre}}(\tau) + \Lambda_{\text{pre}}(\tau)c(\tau), \quad V(\tau) = V_{\text{pre}}^{\lambda(\tau), c(\tau)},$$

and $P(\tau) = P_{\text{pre}}^{\lambda(\tau), c(\tau)}$, gives $\Lambda, X \in AC(J)$ and $a = -\Lambda'$, $b = -X' \in L_{\text{loc}}^1(J)$, and

$$\mathcal{F}(1, 0; V(\tau)) = 0 \text{ for a.e. } \tau \in J.$$

Finally $u = \Lambda^{-1}V(\cdot, \tau) \circ \Phi^{-1}$ by definition of the repaired map, hence the physical identity with $\frac{dt}{d\tau} = \Lambda^2$ follows. $\square$

Theorem 12.2 (Pressure and modulation). *Assume, in addition to Theorem 12.1, that the retained repaired-gauge selection is quantitatively nondegenerate in the sense of Proposition 6.8. Then (V, P) satisfies the renormalized Navier–Stokes equation with the sign convention of Theorem 3.2; there is a decomposition $P = P_{\text{loc}} + H$ on Q_R^* as in Theorem 8.2; and (a, b) is the unique solution of*

$$J(\tau) \begin{pmatrix} a(\tau) \\ b(\tau) \end{pmatrix} = \mathcal{R}(\tau), \quad \begin{pmatrix} a \\ b \end{pmatrix} \in L_{\text{loc}}^1(J)^4,$$

with bounds $|(a, b)| \leq K|\mathcal{R}|$, where K depends on $R, p, \Theta_0, m_0, C_u, \varepsilon_0, \chi_R$.

Proof. Equation and sign convention are inherited directly from Theorem 3.2 applied to (Λ, X) from Theorem 12.1 and the definition of V . By Theorem 8.2 applied at each time slice, the decomposition on Q_R^J is available and unique once $\oint_{B_R} H = 0$ is imposed. The modulation identity is exactly Theorem 9.3; uniqueness follows from invertibility of J (Theorem 6.8) and the pointwise identity $(a, b) = J^{-1}\mathcal{R}$. Uniform bounds on (a, b) are obtained by the bound on J^{-1} there and the stated bound on $\mathcal{R}$. $\square$

Theorem 12.3 (Theorem C: compact-window representation output). *Under the assumptions of Theorems 12.1 and 12.2, for every compact window $J \subset I_*$ and each fixed $R > 0$, setting $Q_R^J := B_R \times J$, the following are available for later papers:*

- (1) the exact renormalized equation on Q_{2R}^J ,
- (2) the four repaired-gauge constraints $\mathcal{F}(1, 0; V(\tau)) = 0$,
- (3) bounds on $a, b \in L_{\text{loc}}^1(J)$ (and L_{loc}^∞ under $\mathcal{R} \in L_{\text{loc}}^\infty$),
- (4) local pressure decomposition $P = P_{\text{loc}} + H$ with $L^{3/2}$ local bounds and harmonic remainder controls,
- (5) renormalized local suitable energy inequality on Q_R^J ,
- (6) stability under subsequences in the topologies above.

No step beyond these items is used in later papers.

Proof. From Theorem 12.1 and the choice of $J \subset I_*$, all quantities are defined on Q_{2R}^J . Items (1) and (2) are Theorems 3.2 and 7.6; item (3) is Theorem 9.3 together with the inverse bound from Proposition 6.8; item (4) is Theorem 8.2; item (5) is Proposition 10.1; item (6) is Proposition 11.1. This is the package exported by the previous theorems. $\square$

Remark 12.4. The results above separate two alternatives cleanly:

- (1) nondegenerate single-core repaired-gauge regime, where Theorems A–C apply;
- (2) gauge degeneracy ($\det J = 0$) or compact-structure failure, which exits the retained stratum and is routed to the multibubble, cascade, or scale-rigidity alternatives.